Welch's t-Test

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Introduction

Welch's t-test is the standard two-sample mean-comparison test when variances across groups may differ. R's `t.test()` defaults to Welch (`var.equal = FALSE`). Decades of simulation evidence favour Welch as the default choice regardless of whether variances appear equal; the efficiency cost versus the equal-variance (Student) test is negligible when variances are equal, and the robustness gain is substantial when they are not.

Prerequisites

Two-sample t-test, variance, degrees of freedom.

Theory

Given independent samples of sizes n_1, n_2 with means $\bar{X}_1, \bar{X}_2$ and sample variances s_1^2, s_2^2, the Welch test statistic is

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}.$$

The Welch-Satterthwaite approximation gives the degrees of freedom

$$\nu = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}.$$

ν is generally not an integer. Under $H_0 : \mu_1 = \mu_2$, $T \sim t_\nu$ approximately.

Assumptions

- Independent observations within and across groups.
- Approximately normal observations (or large n).
- No extreme outliers.

Equal variances are **not** required.

R Implementation

```r
library(effectsize); library(broom)
set.seed(2026)

n1 <- 25; n2 <- 40
group1 <- rnorm(n1, mean = 100, sd = 8)
group2 <- rnorm(n2, mean = 106, sd = 18)   # larger variance

df <- data.frame(
  x     = c(group1, group2),
```

```r
  group = factor(rep(c("A", "B"), c(n1, n2)))
)

# Welch (default)
t.test(x ~ group, data = df) |> tidy()

# Student (var.equal = TRUE) for comparison
t.test(x ~ group, data = df, var.equal = TRUE) |> tidy()

# Effect size
cohens_d(x ~ group, data = df)
```

Output & Results

```
# Welch
estimate1 = 100.8, estimate2 = 106.7, t = -1.77, df = 58.9, p = 0.082

# Student (for comparison)
estimate1 = 100.8, estimate2 = 106.7, t = -1.56, df = 63,  p = 0.124

Cohen's d | 95% CI
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-0.40     | [-0.93, 0.14]
```

Welch's df (58.9) is smaller than Student's (63) because the variances differ; the Welch p-value is consequently smaller in this case, though both fail to reject.

Interpretation

Reporting "Welch's t(58.9) = -1.77, p = 0.082" makes the Welch variant explicit. Cohen's *d* uses a pooled SD by default; for very unequal variances the "Glass's delta" alternative uses only one group's SD.

Practical Tips

- Do not run Levene's test to decide between Welch and Student; a conditional choice inflates Type I error. Default to Welch.
- Satterthwaite df can be fractional; report it as a number with decimals.
- Welch's test is slightly more conservative than Student when variances are equal; the cost is negligible.
- For rank-based alternatives, use Mann-Whitney U.
- Welch-style corrections exist for ANOVA (Welch's F) and linear models (heteroscedasticity-consistent SEs).

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem

- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests